Empriical Studies of Auctions

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- How well do theoretical models of auction explain real-world bidding behavior?
- Can we use auction theory models to guide empirical analysis of bidding markets, e.g.
  - Assess market power or level of competition
  - Identify bidder collusion
  - Evaluate affect of merger, or change in auction rules
  - Assess the importance of asymmetric information
- How can the combination of theory and data guide auction design decisions?

Brief History of Empirical Auction Studies

- Early descriptive work in 1960s and 1970s describing features of bidding for treasury bills, oil leases, timber in national forests.
  - Johnson (1979), Hansen (1986) use change in US Forest Service policy to compare revenue in open and sealed bid auctions — results are inconclusive.
- Hendricks and Porter (1988) use Milgrom-Weber theory of common value auctions with an informed and uninformed bidder to analyze behavior in "drainage tract" oil lease auctions.
  - They show, remarkably, that bidders owning neighboring tracts make much higher expected profit than *de novo* bidders with potentially less information.
  - Athey and Levin (2001) use *ex post* data to test for presence of asymmetric information in timber auctions, and identify gaming of auction rules.

Brief History, cont.

- Paarsch's (1992) stanford dissertation estimates parametric IPV and CV sealed tender models and tests between them.
- Laffont, Ossard and Vuong (1995) show how prices from an ascending auction data can be used to estimate bidder value distributions, apply their idea to french eggplant auctions.
- Guerre, Perrigne and Vuong (2000) show how bid data from sealed bid auctions can be "inverted" to recover bidder value distributions, using an old idea from the IO literature.
- Dozens of papers follow develop and extend this idea to ascending auction data, multi-unit auctions, studies of collusion, market power, etc...

- Bidders 1, ..., N draw independent private values from F.
- In sealed tender, bidder i solves:

$$\max_{b_i} (v_i - b_i) \Pr(\max_{j \neq i} \beta(v_j) \leq b_i) = \max_{b_i} (v_i - b_i) F(\beta^{-1}(b_i))^{N-1}$$

- The equilibrium condition (FOC + symmetric eqm $\beta(\bullet)$):

$$v_i - \beta(s_i) = \frac{1}{\beta'(v_i)} \frac{1}{(N-1) f(v_i) / F(v_i)}$$

- Solving this differential equation:

$$b_i = \beta(v_i) = \frac{\int_{-\infty}^{v_i} x \; f(x) F(x)^{N-2} \, dx}{F(v_i)^{N-1}} = \mathbb{E}[\max_{j \neq i} v_j | v_i \geq \max_{j \neq i} v_j].$$

Relevant Theory: Re-Frame Problem

- Define:

$$G_i(b) = \Pr(\max_{j \neq i} b_j \leq b_i) = \Pr\left(b_i \text{ is winning bid}\right)$$

- Rewrite bidder i's problem is

$$\max_{b_i} \left(v_i - b_i\right) G_i(b)$$

- And in equilibrium, we must have

$$b_i = v_i - \frac{G_i(b_i)}{g_i(b_i)}.$$

- Note the analogy to standard monopoly/monopsony theory: $G_i(b)$ is a *residual supply curve* facing bidder i.

Guerre et al. Estimation Strategy

- Data consist of bids $b_{1t}, ..., b_{Nt}$ from T auctions...
- Fix a bidder i. Use observed bids to construct

$$G_i(b) = \Pr\left(\max_{j \neq i} b_j \leq b_i\right) = \prod_{j \neq i} \Pr(b_j \leq b_i | X_t)$$

- The right hand side can be estimated from the data: $\hat{G}_i$.
- Use equilibrium condition to recover v_i's!

$$v_{it} = b_{it} + \frac{\hat{G}_i(b_i)}{\hat{g}_i(b_i)}.$$

- Does not require bidder symmetry, and can be extended to allow each auction to have different "characteristics" x_t, so $\hat{G}_i(b_i|x_t)$, or to allow for correlated bids/values.

Comparing Open and Sealed Bid Auctions

- Athey, Levin and Seira (2008) paper on timber auctions.
- U.S. Forest Service uses mix of open and sealed bidding
  - 1980s, two distinct regions: Northern, California
  - Tract size and location largely determined format, format was randomized in some sales.
- Under the assumptions of the revenue equivalence theorem:
  - Expect same revenue, participation, allocation on average.
  - Assumptions: symmetry, risk neutrality, independent values, competitive bidding.
- Data: observe: sale format, bids, identities, lots of information about each tract being sold.

Outline of the approach

- Regression analysis shows departures from RET.
- Model to explain departures: relax RET assumptions
  - Heterogeneous bidders: mills v. loggers
  - (Non)-competitive bidding at ascending auctions
- Use GPV method to estimate parameters of model and...
  - Assess whether model can explain RET departures.
  - Assess competitiveness (competition vs. collusion)
  - Welfare analysis of sealed vs open bidding.

Comparing Auction Results

- Estimate OLS regression:

$$Y_t = \alpha \bullet Sealed_t + X_t\beta + N_t\gamma + \varepsilon_t$$

  - Y_t is outcome (entry, revenue, etc.)
  - X_t auction characteristics, N_t "potential" entrants.
  - $Sealed_t$ dummy equal to one if sealed bid.
- Interpretation: α measures "average" difference between running sealed and open auction, holding fixed the sale environment.

Comparing Auction Results

- Each entry interpreted as percentage increase in sealed bid auction relative to open auction

	ln(Logger Entry)	ln(Mill Entry)	Logger Wins	ln(Revenue)
Northern				
	0.089	-0.014	0.039	0.094
	(0.036)	(0.030)	(0.026)	(0.038)
California				
	0.101	-0.026	0.036	0.027
	(0.045)	(0.038)	(0.036)	(0.051)

Asymmetric IPV Model with Costly Entry

- Mills and loggers, all risk-neutral, IPV.
- A bidder must pay K to learn his value and enter.
- Bidder i has private value $v_i \sim F_M(\cdot), F_L(\cdot)$
- Assume F_M is *stochastically higher* than F_L.
- Bidders observe who they're bidding against before submitting bids.
- Let N_M, N_L denote number of potential entrants.
- Let n_M, n_L denote realized entrants.

Asymmetric IPV Auctions

- *Fixed participation (Maskin and Riley, 2000):* In ascending auction, highest-value wins. In first-price auction, mills bid less aggressively than loggers, i.e. $b_M(v) < b_L(v)$ for all v. So loggers win more often and make higher profits in first-price auction relative to ascending auction.
- *Endogenous participation:* Under some conditions, there exists unique type-symmetric equilibrium where: (i) Mills enter with probability one in both formats. (ii) Loggers randomize, enter with higher probability in first-price.
- Basic model predictions
  - Mill bids are higher than logger bids in both formats.
  - Loggers participate and win more in sealed bid auctions.
  - Ambiguous revenue comparison: depends on parameters.
  - Ascending auction is socially efficient; sealed isn't.

Applying the model to data

- Use sealed bid data to estimate model primitives
  - the value distributions F_L, F_M and entry cost K.
- Ask if calibrated model can explain data from both kinds of auctions, and if not, what change in assumptions will allow it to do so.
- Empirical work proceeds in a series of steps.

(1) Estimate bid distributions

- Each sale is different. Conditional bid distributions $G_L(\cdot|X, N, \xi, n)$, $G_M(\cdot|X, N, \xi, n)$.
  - Some differences are observed: X, N, n
  - Some differences aren't observed: $u \sim G_u$.
- Assume $b_{it} = h(X_t) \cdot \phi(u_t) \cdot \eta_{it}$ — implies G_L, G_M, G_U are identified.
- Details: fit Weibull distribution to observed bids:

$$G_k\left(b|X, N, u, n\right) = 1 - \exp\left(-u \cdot \left(\frac{b}{\lambda_k(X, N, n)}\right)^{\rho_k(n)}\right),$$

  - $\ln \lambda_k(X, n) = X\beta_X + N\beta_N + n\beta_n + \beta_k$ and $\ln \rho_k(n) = n\gamma_n + \gamma_k$.
  - assume u is Gamma distributed with mean 1, variance θ.
  - Estimate parameters β, γ, θ by maximum likelihood.

(2) Estimate value distributions

- Equilibrium condition for optimal bidding:

$$b_{it} = v_{it} + \frac{1}{\sum_{j \neq i} \frac{g_j(b_{it}|X_t, N_t, u_t, n_t)}{G_j(b_{it}|X_t, N_t, u_t, n_t)}}$$

- Due to unobservable u, can't recover each v_{it}...
- Instead recover value *distributions* $F_L(\cdot|X, N, u)$, $F_M(\bullet|X, N, u)$

$$F_i(v|X, N, u) = G_i(\beta_i(v)|X, N, u)$$

(3) Calculate expected outcomes given entry

- Have already estimated $F_L(\bullet|X, N, u)$, $F_M(\bullet|X, N, u)$ and $G_u(u)$.
- Given tract characteristics (X, N, u) and entry (n_L, n_M) can calculate sealed bid equilibrium and (easily) open auction equilibrium. Integrating out u gives expected outcomes:
  - Expected sealed and open prices given (X, N, n)
  - Expected sealed and open allocation given (X, N, n)
  - Expected logger and mill profits given (X, N, n)

(4) Estimate distribution of logger entry

- In unique entry equilibrium, all mills enter and each logger enters with probability p_t.
- Specify functional form for p as function of (X, N)

$$p(X, N) = \frac{\exp\left(X\alpha_X + N\alpha_N\right)}{1 + \exp\left(X\alpha_X + N\alpha_N\right)}$$

- Estimate α by maximum likelihood.

(5) Estimate entry costs

- Equilibrium condition for logger entry (randomization):

$$\sum_{n_L} \Pi_L(X, N, n_L, n_M) \cdot \Pr(n_L, n_M | X, N, i \text{ enters}) = K(X, N)$$

- We've estimated $\Pi_L(X, N, n)$, and know $n_M = N_M$.
- Distribution of logger entry is binomial:

$$\Pr(n_L | X, N, i \text{ enters}) = \binom{N_L - 1}{n_L - 1} p(X, N)^{n_L - 1} (1 - p(X, N))^{N_L - n_L}.$$

- Plug in to calculate LHS and infer $K(X, N)$.

(6) Compute expected auction outcomes

- Each sale tract is characterized by a pair (X, N)
- We have estimated $F_L(\cdot|X, N, \xi)$, $F_M(\cdot|X, N, \xi)$, $G(\cdot)$ and $K(X, N)$
- We have computed sealed and open equilibria for all (X, N, ξ, n)
- We have computed expected outcomes given (X, N, n).
- Solve for sealed/open entry equilibria given (X, N).
- Compute expected entry and auction outcomes given (X, N).

(7) Answering economic questions with the model

- How strong are mills compared to loggers?
- How large are profit margins in bids? How large are entry costs?
- How well does the model explain departures from RET?
- Does the assumption of competition or collusion fit the data better?
- How important is endogenous entry?
- What are the welfare consequences of open vs sealed bidding?

Estimation results

- Bids increase with number of competitors.
- Mill bids are 15-25% greater than logger bids on average.
- Mills bid larger margins: mill bid function lies below logger's.
- Estimated profit margin (cond'l on entry): 10.7% at the median.
- Estimated entry cost: $4695 at the median.

Predicted versus Actual Outcomes

	Actual	Predicted Cond'l on Entry		Predicted w/ Eqm Entry	
Sealed Bid Sales *(in-sample predictions)*					
Avg. Sale Price	69.4	69.9	(1.4)	70.4	(1.6)
Logger Wins (%)	68.1	68.0	(0.9)	65.0	(0.01)
Logger Entry	3.23			3.23	(0.1)
Open Sales *(out-of-sample predictions)*					
Avg. Price *(competition)*	63.3	67.9	(1.8)	67.8	(2.1)
Avg. Price *(collusion)*	63.3	44.2	(1.3)	44.1	(2.2)
Logger Wins (%)	60.0	56.0	(0.01)	54.4	(0.02)
Logger Entry	2.75			2.67	(0.2)

Welfare Comparison: Sealed vs. Open

	Sealed vs. Competitive Open Auctions		Sealed vs. Part. Collusive Open Auctions	
Exogenous Entry				
Avg. Sale Price	0.03%	(0.04)	3.98%	(0.24)
Avg. Sale Surplus	-0.08%	(0.02)	same as comp.	
Logger Wins	0.46%	(0.16)	same as comp.	
Predict Entry & Bidding				
Avg. Sale Price	1.49%	(0.71)	5.57%	(0.82)
Avg. Sale Surplus	-0.30%	(2.70)	same as comp.	
Logger Wins	2.46%	(0.00)	same as comp.	
Logger Entry	0.34%	(0.12)	same as comp.	

Other interesting auction work

- Hortacsu (2002) uses T-bill auction data to estimate values of bidders and compare uniform versus discriminatory price auctions.
- Wolak (2003, etc.) uses electricity auction data to estimate cost of generation units and study aspects of electricity market design, including competitiveness.
- Haile and Tamer (2003) show how data from *open* auctions can be used to provide *bounds* on bidder values, though not exact estimates without strong assumptions about bidding behavior.
- Hendricks, Pinske and Porter (2000) estimate a *common value* model of oil lease auction and use it to calculate the relevant information contained in competing bids.